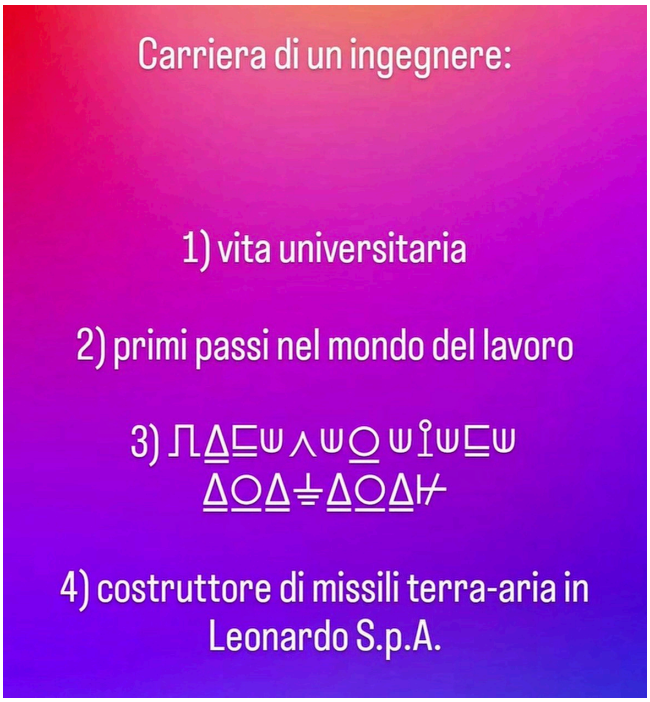
Technische Universität Wien

Computer-Aided Verification

course material: slides



meme translation: “an engineer’s carreer: 1) uni life 2) first steps in the job market 3) [whatever that is] 4) surface-to-air missile builder for Leonardo S.p.A (famous italian defense company with very dubious ethics)”

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CHAPTER 1

Introduction

1.1 | Verification & Validation

Software and integrated circuits are pervasive, and a huge effort is spent on verification and validation (V&V). Approximately 50% of software development time and 35% of hardware development time is dedicated to these tasks. The core goal of Computer-Aided Verification (CAV) is either to formally establish the correctness of a system or to find bugs.

1.2 | Model Checking Basics

Def. 1 **Transition Systems**

Programs and circuits are modeled as **transition systems**.

- **States (S):** represent configurations, such as program counter values (pc) and variable assignments.
- **Transitions (T):** define how states change, essentially capturing the operational semantics of the system by mapping current states to next states ($s \to s'$).

A formal model checking problem can consist of:

- starting states (i): the initial configurations of the system;
- transitions (t): the allowable state updates;
- bad states ($\neg p$): states that violate the specification;

Violations of specifications are witnessed by a **counterexample**.

Safety vs Liveness violations

- **Safety violation:** a counterexample consisting of a path of *finite length* that reaches a bad state.
  - example: ”two intersecting traffic lights must never be green at the same time”
    counterexample: a finite path where the North-South light turns green, a timer glitches, and the East-West light also turns green - the path ends the moment this bad state is reached
- **Liveness violation:** a counterexample consisting of a path of *infinite length* that repeatedly visits bad states and never reaches a good state.
  - example: ”if a pedestrian presses the crosswalk button, the walk signal will eventually turn green”

counterexample: an infinite loop where the pedestrian presses the button, but the traffic lights never turn green

To properly define ”bad” or ”good” states, engineers verify systems against formal specifications written in **Temporal Logic**. This replaces natural language requirements with precise mathematical constraints (e.g., $GFp \rightarrow G(Gq \vee Fr)$).

1.3 | The State-Space Explosion Problem

The main challenge in model checking is the exponential growth of possible states.

Sources of state explosion

- **Data:** enumerating large domains of values, including user inputs;
- **Concurrency:** enumerating all possible process and thread interleaving schedules;
- **Environment:** enumerating all possible states of the execution environment;
- **Faults:** faulty processes that may feed others with junk data;

1.3.1 | Remedies

To mitigate state-space explosion, several techniques are utilized:

- **Abstraction:** mapping concrete values into a coarser, smaller domain;
- **Symbolic techniques:** using Binary Decision Diagrams (BDDs), boolean satisfiability solvers (SAT), and satisfiability-modulo-theory (SMT) solvers to handle sets of states;
- **Partial order reduction:** avoiding the exploration of states via ”redundant paths”;
- **Symmetry reduction:** avoiding the exploration of symmetric states;

1.4 | Symbolic Model Checking

Instead of exploring states one by one (explicit state model checking), symbolic model checking processes sets of states simultaneously.

- **Sets of States** are epresented as *logical formulae* $F(V)$ over program variables, registers, and signals
- **Transitions** are represented as *binary relations* $T(V, V')$ that link current variables V to next-state variables V'

example: reachability

Reachability can be mathematically expressed to check if a property P can be violated in k steps:

$$I(V_0) \wedge \left(\bigwedge_{i=1}^{k} T(V_{i-1}, V_i) \right) \wedge \neg P(V_k)$$

where

- $I(V_0)$ means that V_0 is an initial state
- $\bigwedge_{i=1}^{k} T(V_{i-1}, V_i)$ represents the (valid) transitions
- $\neg P(V_k)$

1.4.1 | Invariants and Fixpoints

When computing reachable states iteratively, we define $R_0 = I$, and subsequent reachable sets as $T(R_{i-1}) = R_i$. The set of all states reachable in up to k steps is $R_{\leq k} = \bigcup_{i=0}^{k} R_i$.

Def. 2 **Safety condition**

A system is considered safe if:

1 | $R_{\leq k}$ contains the initial states I.

2 | T cannot leave $R_{\leq k}$ (the system has reached a **fixpoint**).

3 | $R_{\leq k}$ does not overlap with the bad states $\neg P$.

Since exact reachability computations retard convergence and require complex quantifier alternation ($\forall\exists$), practical approaches often compute over-approximate inductive frames to detect fixpoints more efficiently.

1.5 | Abstraction and CEGAR

Abstraction reduces the state space but introduces non-determinism and may introduce divergent behaviors (non-termination). An abstract program must strictly **over-approximate** the original program, meaning it contains ”more” behaviors than the concrete system.

To handle this automatically, modern model checkers use the **Counterexample-Guided Abstraction Refinement (CEGAR)** loop:

1 | **check abstraction:** see if the abstract model is safe; if safe, the concrete system is safe

2 | **check feasibility:** if a counterexample is found, check if it is feasible in the concrete system; if feasible, a real bug is found

3 | **refine:** if the counterexample is infeasible (a false positive caused by over-approximation), use the failure trace to refine the abstraction and check again

1.6 | Protocols & Execution Models

Verification models differ based on whether they target hardware or software.

- **Concurrent Protocols (software):** standard assumptions include that processes run **asynchronously** (interleave), the scheduler is completely non-deterministic, and individual statements execute atomically; examples include threads accessing shared variables (requiring mutual exclusion algorithms like Peterson’s)
- **Hardware Protocols:** standard assumptions include that components run **synchronously** based on a global clock (every component makes a step at a time) and communicate via actions

or shared memory; examples include cache coherence protocols like MSI (Modified-Shared-Invalid)

CHAPTER 2

Temporal Logics and Model Checking

Def. 3 Finite-State Transition System

A triple (S, T, I) is a **Finite-State Transition System** if

- S is a finite set of states
- $I \subseteq S$ is a set of initial states
- $T \subseteq S \times S$ is a *total* transition relation ($\forall s \in S. \exists s' \in S. T(s, s')$)

Def. 4 Kripke Structure

A quadruple (S, T, I, L) is a **Kripke Structure** if

- (S, T, I, L) is a finite-state transition system
- L is a labelling function $S \to 2^{AP}$(from states to atomic propositions)

Atomic propositions are used to represent properties of states, which can be combined them to generate *assertions*.

We can check whether an assertion holds in a given state via the standard $\vDash$ relation - we write $s \vDash F$ if F holds in state s:

- $s \vDash p \iff p \in L(s)$
- $s \nvDash F \iff p \notin L(s)$ (!! closed world assumption - anything not explicitly defined as true in $L(s)$ is evaluated as false)
- $s \vDash F_1 \vee F_2 \iff s \vDash F_1$ or $s \vDash F_2$
- $s \vDash F_1 \wedge F_2 \iff s \vDash F_1$ and $s \vDash F_2$

Assertions can be used to express that something bad is not supposed to happen (*safety* properties).

For example, if we had a traffic light system, we could write $\texttt{assert}(\neg \texttt{green}_1 \vee \neg \texttt{green}_2)$ to signify that traffic lights 1 and 2 cannot be green at the same time.

2.1 | Explicit-State Model Checking of Assertions

To algorithmically verify if a system design is correct, we can use **explicit-state model checking**. This technique systematically enumerates all reachable states of a finite-state transition system, starting from the initial states I and following the transition relation T, to verify whether a specified property or assertion F holds in every reachable state. The two primary search strategies used to explicitly traverse and evaluate this state space are Depth-First Search (DFS) and Breadth-First Search

(BFS).

Depth-First Search (DFS)
Algorithm ExplicitReachDFS

```
1: open ← list(I)
2: visited ← ∅
3: while open ≠ [] do
4:     s ← head(open)
5:     open ← tail(open)
6:     if s⊭F then
7:         error(s)
8:     end if
9:     for each s′ ∈ S : (s, s′) ∈ T do
10:        if s′ ∉ visited then
11:            visited ← {s′} ∪ visited
12:            open ← s′ :: open          ▷ prepend
13:        end if
14:    end for
15: end while
```

Breadth-First Search (BFS)
Algorithm ExplicitReachBFS

```
1: open ← list(I)
2: visited ← ∅
3: while open ≠ [] do
4:     s ← head(open)
5:     open ← tail(open)
6:     if s⊭F then
7:         error(s)
8:     end if
9:     for each s′ ∈ S : (s, s′) ∈ T do
10:        if s′ ∉ visited then
11:            visited ← {s′} ∪ visited
12:            open ← append(open, s′)          ▷ append
13:        end if
14:    end for
15: end while
```

- keeps only the current stack ($|\text{open}| < |S|$): considered more memory-efficient
- counterexamples can be quite long
- queue grows very large due to frontier size: consumes significantly more memory
- always finds the shortest counterexample

But assertions only allow us to make sure that nothing bad happens: what if we want to guarantee that something *good* happens?

We need to extend our specification language to consider **temporal logics**.

2.2 | Temporal Logics

Temporal logics allow us to reason about properties over paths.

> Def. 5 **Path formulae**
>
> An (infinite) **path** π is a sequence of states $s_0, s_1, \ldots$ such that $T(s_i, s_{i+1})$ for $i \geq 0$.
>
> If we fix a Kripke structure $\mathcal{M}$ and a path π, we can introduce *path formulae* of the form:
>
> $$\mathcal{M}, \pi \vDash \varphi$$
>
> (meaning that φ holds on the path π)

The convention is to denote state formulae by F, and path formulae by φ.

We can use the path semantics for state formulae as well, and say that $\mathcal{M}, \pi \vDash F \iff F$ holds in the first state s_0 of π.

We now introduce a temporal logic called **CTL*** (Computation Tree Logic*), which allows us to reason about computation trees and branching time.

Def: CTL*

CTL* formulae are divided into state and path formulae:

$$F ::= \text{true} \mid p \mid F_1 \wedge F_2 \mid \neg F \mid \mathrm{E}\varphi \mid \mathrm{A}\varphi$$

$$\varphi ::= F \mid \varphi_1 \wedge \varphi_2 \mid \neg\varphi \mid \mathrm{X}\varphi \mid \varphi_1 \mathrm{U} \varphi_2$$

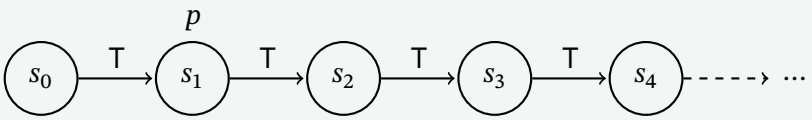
Def. 6 Temporal Operator: Next (X)

Syntax: unary path formula $X\varphi$.

Semantics: $\mathcal{M}, \pi \vDash X\varphi$ if and only if $\mathcal{M}, \pi^1 \vDash \varphi$.

Example

$\mathcal{M}, \pi \vDash Xp$; it does not matter whether or not p holds in s_0 or $s_2, s_3, \ldots$.



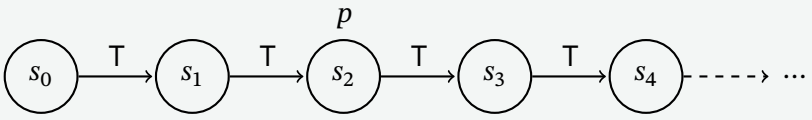
Def. 7 Temporal Operator: Eventually (F)

Syntax: unary path formula $F\varphi$.

Semantics: $\mathcal{M}, \pi \vDash F\varphi$ if and only if $\mathcal{M}, \pi^k \vDash \varphi$ for some $k \geq 0$. Intuitively, p holds after a finite number of steps. F allows us to express basic liveness properties.

Example

$\mathcal{M}, \pi \vDash Fp$.



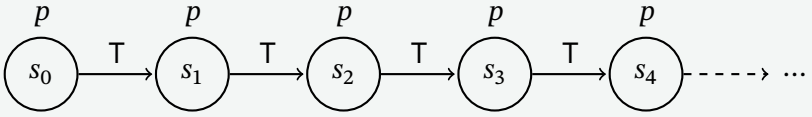
Def. 8 Temporal Operator: Globally (G)

Syntax: unary path formula $G\varphi$.

Semantics: $\mathcal{M}, \pi \vDash G\varphi$ if and only if $\mathcal{M}, \pi^i \vDash \varphi$ for $i \geq 0$. Intuitively, p holds in every path state. G allows us to express basic safety properties.

Example

$\mathcal{M}, \pi \vDash Gp$.



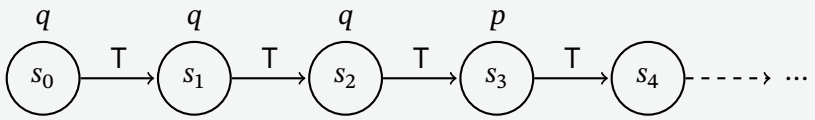
Def. 9 **Temporal Operator: Until (U)**

Syntax: binary path formula $\varphi_1 U \varphi_2$.

Semantics: $\mathcal{M}, \pi \vDash \varphi_1 U \varphi_2$ if and only if there is $k \geq 0$ such that $\mathcal{M}, \pi^k \vDash \varphi_2$ and $\mathcal{M}, \pi^j \vDash \varphi_1$ for $0 \leq j < k$. Intuitively, φ_1 holds until φ_2 holds, and importantly, φ_2 must happen eventually.

Example

$\mathcal{M}, \pi \vDash qUp$. ($q$ doesn't have to hold anymore once discharged by p).



Def. 10 **Temporal Operator: Release (R)**

Syntax: binary path formula $\varphi_1 R \varphi_2$.

Semantics: $\mathcal{M}, \pi \vDash \varphi_1 R \varphi_2$ if and only if one of the two conditions holds:

1 | $\exists k \geq 0$ such that $\mathcal{M}, \pi^k \vDash \varphi_1$ and $\mathcal{M}, \pi^j \vDash \varphi_2$ for $0 \leq j \leq k$. (φ_1 releases φ_2, if φ_1 ever holds).

2 | $\mathcal{M}, \pi^j \vDash \varphi_2$ for $j \geq 0$.

Example

$\mathcal{M}, \pi \vDash pRq$.

